

SCHMITT TRIGGER

Aim:

To design a Schmitt trigger for a specified LTP and UTP and to plot the hysteresis curve by applying ac.

Components required:

CRO, function generator, op-amp, resistors, diodes, power supplies and bread board.

Theory:

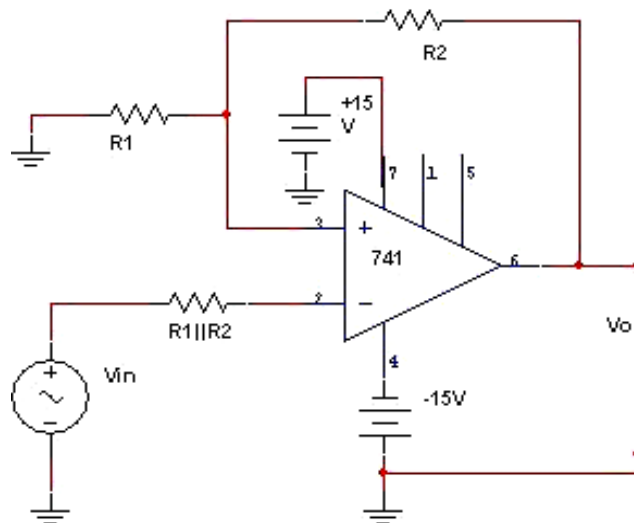
Schmitt trigger is a comparator and it is known as a squaring circuit because it converts an irregular shaped waveform to a square wave. The output voltage changes state every time when the input voltage crosses the thresholds levels.

The input voltage at which the output switches from $+V_{sat}$ to $-V_{sat}$ is called the Upper Triggering Point or Upper Threshold Point(UTP). The input voltage at which the output switches from $-V_{sat}$ to $+V_{sat}$ is called the Lower Triggering Point or Lower Threshold Point(LTP).

These threshold voltages are obtained by the voltage divider R_1 - R_2 . Suppose the output voltage is $+V_{sat}$. Now the voltage across R_1 is $V_{UTP} = \frac{+V_{sat} R_1}{R_1 + R_2}$. When the input voltage exceeds the voltage across the resistor R_1 , the output goes to $-V_{sat}$. Now the voltage across R_2 is $V_{LTP} = \frac{-V_{sat} R_1}{R_1 + R_2}$. When the input voltage goes lower than this voltage, the output goes to $+V_{sat}$.

SCHMITT TRIGGER (USING IC 741):

Procedure:



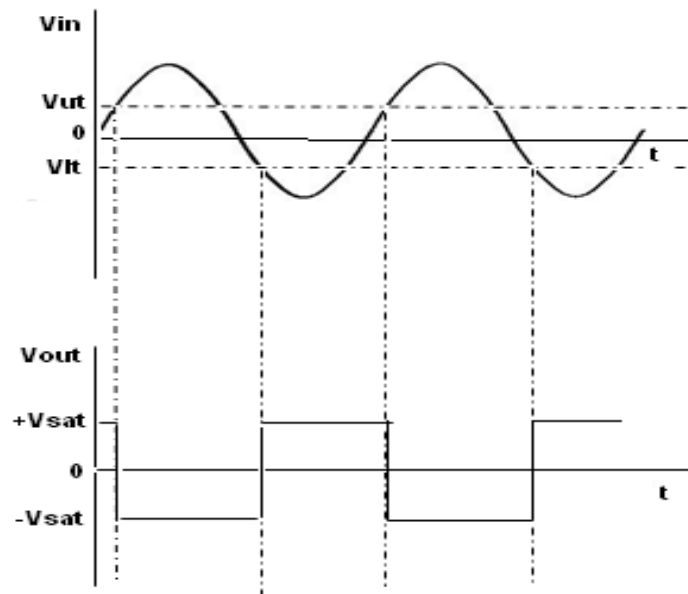
1. Verify whether the op-amp IC is in good condition. This can be done by using it as a voltage follower.
2. Set up the circuits one by one. Observe the input and the output on the CRO screen.
3. Design the circuit for different values of LTP and UTP.
4. Observe the waveform and the transfer characteristics.
5. Draw the hysteresis curve.

Determine Upper threshold and Lower threshold voltages using the formulae

$$V_{UT} = \frac{R_1}{R_1 + R_2} (+v_{sat})$$

$$V_{LT} = \frac{R_1}{R_1 + R_2} (-v_{sat})$$

Observations:



<i>Voltage (V)</i>	<i>Designed (V)</i>	<i>Measured (V)</i>
V_{UT}		
V_{LT}		